



Date: _____

TUTORIAL -1 SDLC & STLC PHASES

Aim: - Understand SDLC and STLC Phases for a Real-World Software Project and Identify Testing Activities Performed During Each Development Stage.

Description :

Problem Statement :

Implementation Details :

Parameter	Details
Project Name	
Application/Software	
Module Name	
SDLC Model	
Programming Language	
IDE/Tool Used	
Input/Test Data	

Source Code :



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Test Case Detailed :

Field	Details
-------	---------



Test Case ID	
Requirement ID	
Module Name	
Feature Under Test	
Test Scenario	
Test Objective	
Test Type (Unit/Integration/System/UAT)	
Test Technique (Black Box/White Box)	
Priority	
Severity	
Preconditions	
Postconditions	
Test Environment	
Browser/OS (if applicable)	
Test Data	
Expected Result	
Actual Result	
Status (Pass/Fail)	
Executed By	
Execution Date	

Test Case Execution :

Step No.	Test Steps	Input	Expected Result	Actual Result	Status



Testing Activities Performed :

Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable/Artifact
1				
2				
3				
4				
5				
6				

Observation :

Sr. No.	Observation	Remarks
1		
2		



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3		
4		
5		

Output :

Program Outcome :



Solve the below mention questionnaires:

1. What does SDLC stand for?

- A) Software Design Life Cycle
- B) Software Development Life Cycle
- C) System Design Life Cycle
- D) Software Deployment Life Cycle

Answer:

2. Which is the first phase of SDLC?

- A) Testing
- B) Deployment
- C) Requirement Analysis
- D) Maintenance

Answer:

3. What does STLC stand for?

- A) Software Technology Life Cycle
- B) Software Testing Life Cycle
- C) System Testing Life Cycle
- D) Software Test Logic Cycle

Answer:

4. Which STLC phase involves preparing test cases?

- A) Requirement Analysis
- B) Test Planning
- C) Test Case Development
- D) Test Closure

Answer:

5. Which testing is performed by developers during coding?

- A) System Testing
- B) Integration Testing
- C) Unit Testing
- D) Acceptance Testing

Answer:



Conclusion / Outcome:

Rubrics:

Evaluation Criteria	Maximum Marks	Marks Obtained
Description & Problem Statement	2	
Implementation & Source Code	3	
Test Cases, Execution & Output	2	
Observation & Conclusion	2	
Documentation	1	
Total	10	

Signature with Date of Completion	
Marks out of 10	

Date: _____

TUTORIAL -2 TEST PLAN & RTM

Aim: - Understand a professional test plan, Requirement Traceability Matrix (RTM), testing scope document, and testing strategy for a sample application.

Description :



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Problem Statement :

Implementation Details :

Parameter	Details
Project Name	
Application/Software	
ModuleName	
SDLC Model	
Programming Language	
IDE/Tool Used	
Input/Test Data	

Source Code :



Test Case Detailed :

Field	Details
Test Case ID	
Requirement ID	
Module Name	
Feature Under Test	
Test Scenario	
Test Objective	
Test Type (Unit/Integration/System/UAT)	
Test Technique (Black Box/White Box)	
Priority	
Severity	

Preconditions	
Postconditions	
Test Environment	
Browser/OS (if applicable)	
Test Data	
Expected Result	
Actual Result	
Status (Pass/Fail)	
Executed By	
Execution Date	

Test Case Execution :

[illegible]



Testing Activities Performed :

Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable/Artifact
1				
2				
3				
4				
5				
6				
7				

Observation :

Sr. No.	Observation	Remarks
1		
2		



3		
4		
5		

Output :

Solve the below mention questionnaires:

1. What is the primary purpose of a Test Plan?

- A) Coding Standards
- B) Testing Management
- C) Database Design



D) Deployment

Answer:

2. What does RTM stand for?

- A) Requirement Technical Matrix
- B) Requirement Traceability Matrix
- C) Resource Tracking Matrix
- D) Requirement Test Matrix

Answer:

3. What is the main purpose of RTM?

- A) To create source code
- B) To map requirements with test cases
- C) To design the database
- D) To deploy software

Answer:

4. Which document defines functionalities that are included and excluded from testing?

- A) User Manual
- B) Testing Scope Document
- C) Source Code
- D) Design Specification

Answer:

5. Which document defines the overall testing methodology and approach?

- A) Bug Report
- B) RTM
- C) Testing Strategy
- D) User Guide

Answer:

Conclusion / Outcome:

Rubrics:

Evaluation Criteria	Maximum Marks	Marks Obtained
Description & Problem Statement	2	



Implementation & Source Code	3	
Test Cases, Execution & Output	2	
Observation & Conclusion	2	
Documentation	1	
Total	10	

Signature with Date of Completion	
Marks out of 10	

Date: _____

TUTORIAL -3 Requirement Reviews & Inspections

Aim: - Understand Requirement Reviews, Code Walkthroughs, Inspections, and Peer Review Activities Using Sample Project Documents.

Description :

Problem Statement :

Implementation Details :

Parameter	Details
Project Name	
Application/Software	



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Module Name	
SDLC Model	
Programming Language	
IDE/Tool Used	
Input/Test Data	

Source Code :



Test Case Detailed :

Field	Details
Test Case ID	
Requirement ID	
Module Name	
Feature Under Test	
Test Scenario	
Test Objective	
Test Type (Unit/Integration/System/UAT)	
Test Technique (Black Box/White Box)	
Priority	



Severity	
Preconditions	
Postconditions	
Test Environment	
Browser/OS (if applicable)	
Test Data	
Expected Result	
Actual Result	
Status (Pass/Fail)	
Executed By	
Execution Date	

Test Case Execution :

Step No.	Test Steps	Input	Expected Result	Actual Result	Status



Testing Activities Performed:

Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable/Artifact
1				
2				
3				
4				
5				

Observation :

Sr. No.	Observation	Remarks
1		
2		
3		
4		
5		



Output :

Solve the below mention questionnaires:

1. Which testing technique identifies defects without executing the software?

- A) Dynamic Testing
- B) Static Testing
- C) Regression Testing
- D) Performance Testing

Answer:

2. Which document is primarily reviewed during Requirement Review?

- A) Source Code
- B) Database Schema
- C) Software Requirement Specification (SRS)
- D) Test Report

Answer:

3. What is the primary purpose of a Code Walkthrough?

- A) Execute software
- B) Deploy software
- C) Review code and identify defects
- D) Create database tables

Answer:

4. Inspection is a _____ review process.



- A) Informal
- B) Dynamic
- C) Automated
- D) Formal

Answer:

5. Peer Review is performed by:

- A) Customers
- B) End Users
- C) Team Members Reviewing Each Other's Work
- D) System Administrators

Answer:

Conclusion / Outcome:

Rubrics:

Evaluation Criteria	Maximum Marks	Marks Obtained
Understanding of Requirement Reviews, Code Walkthroughs, Inspections, and Peer Review Concepts	2	
Analysis of Sample Project Documents (SRS, Source Code, Review Artifacts)	2	
Preparation of Test Cases, Review Findings, and Testing Activities	3	
Observation, Defect Identification, and Conclusion	2	
Documentation, Presentation, and Neatness	1	
Total	10	



Signature with Date of Completion	
Marks out of 10	

Date: _____

TUTORIAL -4 Functional & Non-Functional Testing

Aim: - Apply Functional, Non-Functional, Positive, and Negative Test Cases for Modules Such as Login, Registration, and Payment Systems.

Description :

Problem Statement :

Implementation Details :

Parameter	Details
Project Name	
Application/Software	
Module Name	
SDLC Model	
Programming Language	
IDE/Tool Used	



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Input/Test Data	
-----------------	--

Source Code :



Test Case Detailed :

Field	Details
Test Case ID	
Requirement ID	
Module Name	
Feature Under Test	
Test Scenario	
Test Objective	
Test Type (Unit/Integration/System/UAT)	
Test Technique (Black Box/White Box)	
Priority	
Severity	
Preconditions	
Postconditions	
Test Environment	
Browser/OS (if applicable)	
Test Data	
Expected Result	
Actual Result	
Status (Pass/Fail)	
Executed By	
Execution Date	

Test Case Execution :



Step No.	Test Steps	Input	Expected Result	Actual Result	Status

Testing Activities Performed:

Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable/Artifact
1				
2				
3				
4				
5				

Observation :

Sr. No.	Observation	Remarks
---------	-------------	---------



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1		
2		
3		
4		
5		

Output :



Solve the below mention questionnaires:

1. Which testing verifies software functionality against requirements?

- A) Non-Functional Testing
- B) Functional Testing
- C) Security Testing
- D) Performance Testing

Answer:

2. Positive Testing is performed using:

- A) Invalid Inputs
- B) Incorrect Data
- C) Valid Inputs
- D) Random Inputs

Answer:

3. Which testing verifies system behavior with invalid inputs?

- A) Positive Testing
- B) Regression Testing
- C) Integration Testing
- D) Negative Testing

Answer:

4. Performance Testing is a type of:

- A) Functional Testing
- B) Non-Functional Testing
- C) Unit Testing
- D) Acceptance Testing

Answer:

5. Which module is commonly tested for valid username and password verification?

- A) Payment Module
- B) Registration Module
- C) Login Module
- D) Billing Module

Answer:



Conclusion / Outcome:

Rubrics:

Evaluation Criteria	Maximum Marks	Marks Obtained
Understanding of Functional, Non-Functional, Positive, and Negative Testing Concepts	2	
Test Case Design for Login, Registration, and Payment Modules	3	
Test Case Execution, Results, and Output	2	
Observation, Defect Identification, and Conclusion	2	
Documentation, Presentation, and Neatness	1	
Total	10	

Signature with Date of Completion	
Marks out of 10	

Date:_____

TUTORIAL -5 Defect Reporting

Aim: - Apply defect identification and prepare professional bug reports including severity, priority, screenshots, and reproduction steps.

Description :



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Problem Statement :

Implementation Details:

Parameter	Details
Project Name	
Application/Software	
Module Name	
Version	



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Programming Language	
IDE/Tool Used	
Database Used	
Operating System	
Browser (If Applicable)	
Test Environment	
Tester Name	
Test Date	

Source Code :



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Bug Identification Details :

Field	Details
Bug ID	
Defect Title	
Module Name	
Feature Affected	
Defect Description	
Defect Type	



Severity	
Priority	
Environment	
Reported By	
Date Reported	

Steps to Reproduce :

Step No.	Action Performed	Input/Test Data	Expected Result	Actual Result

Bug Report Template :

Field	Details
Bug ID	
Bug Summary	
Module Name	
Severity	
Priority	



Steps to Reproduce	
Expected Result	
Actual Result	
Assigned To	
Status(New/Open/InProgress/Fixed/Closed)	
Resolution	
Retest Status	
Remarks	

Bug Summary :

Bug ID	Module	Severity	Priority	Status



Observation :

Sr. No.	Observation	Remarks

Output :

Solve the below mention questionnaires:

1. What is a software defect?

- A) A new software feature
- B) A problem that causes the software to behave incorrectly
- C) A software update
- D) A testing tool

Answer:

2. Which document is used to record and track software defects?



- A) User Manual
- B) Test Plan
- C) Bug (Defect) Report
- D) Project Schedule

Answer:

3. Severity in a bug report indicates:

- A) The order in which the defect should be fixed
- B) The impact of the defect on the system
- C) The name of the tester
- D) The number of test cases executed

Answer:

4. Which section of a bug report helps developers reproduce the defect?

- A) Module Name
- B) Steps to Reproduce
- C) Resolution
- D) Priority

Answer:

5. Why are screenshots included in a bug report?

- A) To increase the size of the report
- B) To provide visual evidence of the defect
- C) To improve software performance
- D) To reduce the number of test cases

Answer:

Conclusion / Outcome:

Rubrics:

Evaluation Criteria	Maximum Marks	Marks Obtained
Understanding of Defect Identification and Bug Reporting Concepts	2	



Defect Identification and Preparation of Professional Bug Reports	3	
Severity, Priority, Reproduction Steps, and Supporting Evidence (Screenshots)	2	
Observation, Defect Analysis, and Conclusion	2	
Documentation, Presentation, and Neatness	1	
Total	10	

Signature with Date of Completion	
Marks out of 10	

Date: _____

TUTORIAL -6 Manual Test Execution

Aim: - Apply manual test cases using multiple input conditions and prepare detailed test execution and summary reports.

Description :

Problem Statement :

Implementation Details :

Parameter	Details
-----------	---------



Project Name	
Application/Software	
Module Name	
SDLC Model	
Programming Language	
IDE/Tool Used	
Input/Test Data	

Source Code :



Test Case Detailed :

Field	Details
Test Case ID	
Requirement ID	
Module Name	
Feature Under Test	
Test Scenario	
Test Objective	
Test Type (Unit/Integration/System/UAT)	
Test Technique (Black Box/White Box)	
Priority	
Severity	
Preconditions	
Postconditions	



Test Environment	
Browser/OS (if applicable)	
Test Data	
Expected Result	
Actual Result	
Status (Pass/Fail)	
Executed By	
Execution Date	

Test Case Execution :

Step No.	Test Steps	Input	Expected Result	Actual Result	Status

Testing Activities Performed:

Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable/Artifact
1				



2				
3				
4				
5				
6				

Observation :

Sr. No.	Observation	Remarks
1		
2		
3		
4		
5		

Output :



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Solve the below mention questionnaires:

1. What is Manual Testing?

- A) Testing using automation tools
- B) Testing performed manually without automation tools
- C) Database Testing
- D) Network Testing

Answer:

2. Which document records actual results of executed test cases?

- A) Design Document
- B) Test Execution Report
- C) User Manual
- D) RTM



Answer:

3. What is compared during test execution?

- A) Source Code and Design Document
- B) Expected Result and Actual Result
- C) Database and UI
- D) Hardware and Software

Answer:

4. Which input condition checks validation handling?

- A) Valid Username and Password
- B) Blank Input Values
- C) Correct Login Credentials
- D) Successful Transaction

Answer:

5. What is the purpose of a Test Summary Report?

- A) Writing Source Code
- B) Designing Database Tables
- C) Providing Overall Testing Status and Results
- D) Creating User Interfaces

Answer:

Conclusion / Outcome:

Rubrics:

Evaluation Criteria	Maximum Marks	Marks Obtained
Understanding of Manual Testing and Test Execution Process	2	
Execution of Manual Test Cases Using Multiple Input Conditions	3	
Test Execution Results, Defect Recording, and Summary Report	2	
Observation, Analysis, and Conclusion	2	
Documentation, Presentation, and Neatness	1	
Total	10	



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Signature with Date of Completion	
Marks out of 10	

Date: _____

TUTORIAL -7 BVA & Equivalence Partitioning

Aim: - Apply Boundary Value Analysis (BVA) and Equivalence Partitioning techniques for validation testing of forms and authentication modules.

Description :

Problem Statement :



Implementation Details:

Parameter	Details
Project Name	
Application/Software	
Module Name	
Version	
Programming Language	
IDE/Tool Used	
Database Used	
Operating System	
Browser (If Applicable)	
Test Environment	
Tester Name	
Test Date	

Source Code :



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Equivalence Class Table :



Input Field	Input Range / Condition	Equivalence Class Type (Valid / Invalid)	Test Value	Expected Result

Boundary Value Analysis (BVA) Table

Input Field	Minimum Value	Just Above Minimum	Nominal Value	Just Below Maximum	Maximum Value	Just Above Maximum

Alternative BVA Format (Industry Standard)



Input Field	Boundary Condition	Boundary Value	Expected Result	Actual Result	Status
	Minimum - 1				
	Minimum				
	Minimum + 1				
	Maximum - 1				

Bug Identification Details :

Field	Details
Bug ID	
Defect Title	
Module Name	
Feature Affected	
Defect Description	
Defect Type	
Severity	
Priority	
Environment	
Reported By	
Date Reported	

Steps to Reproduce :



Step No.	Action Performed	Input/Test Data	Expected Result	Actual Result

Bug Report Template :

Field	Details
Bug ID	
Bug Summary	
Module Name	
Severity	
Priority	
Steps to Reproduce	
Expected Result	
Actual Result	
Assigned To	
Status (New/Open/In Progress/Fixed/Closed)	
Resolution	
Retest Status	
Remarks	

Bug Summary :

Bug ID	Module	Severity	Priority	Status
--------	--------	----------	----------	--------



Observation :

Sr. No.	Observation	Remarks

Output :



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Solve the below mention questionnaires:

1. Equivalence Partitioning is a type of:

- A) White Box Testing
- B) Black Box Testing
- C) Performance Testing
- D) Security Testing

Answer:

2. What is the main objective of Equivalence Partitioning?

- A) Increase number of test cases
- B) Reduce test cases while maintaining coverage
- C) Improve coding standards
- D) Increase execution time

Answer:

3. Boundary Value Analysis focuses on:

- A) Random Values
- B) Internal Code Structure
- C) Boundary Values of Input Data
- D) Database Design

Answer:

4. For a username length range of 5–15, which value is just below the minimum boundary?

- A) 3
- B) 4
- C) 5
- D) 6

Answer:

5. Which testing technique divides input data into valid and invalid groups?

- A) Regression Testing
- B) Integration Testing
- C) Equivalence Partitioning
- D) Acceptance Testing

Answer:

Conclusion / Outcome:



Rubrics:

Evaluation Criteria	Maximum Marks	Marks Obtained
Understanding of Boundary Value Analysis (BVA) and Equivalence Partitioning Concepts	2	
Design of Test Cases Using BVA and Equivalence Partitioning Techniques	3	
Test Execution, Validation Results, and Output	2	
Observation, Analysis, and Conclusion	2	
Documentation, Presentation, and Neatness	1	
Total	10	

Signature with Date of Completion	
Marks out of 10	

Date: _____

TUTORIAL -8 Bugzilla & TestLink

Aim: - Apply defect management, test case management, and execution report generation using Bugzilla and TestLink workflows.

Description :



Problem Statement :

Implementation Details :

Parameter	Details
Project Name	
Application/Software	
Module Name	
SDLC Model	
Programming Language	
IDE/Tool Used	
Input/Test Data	

Source Code :



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Test Case Detailed :

Field	Details
Test Case ID	
Requirement ID	
Module Name	
Feature Under Test	
Test Scenario	
Test Objective	
Test Type (Unit/Integration/System/UAT)	



Test Technique (Black Box/White Box)	
Priority	
Severity	
Preconditions	
Postconditions	
Test Environment	
Browser/OS (if applicable)	
Test Data	
Expected Result	
Actual Result	
Status (Pass/Fail)	
Executed By	
Execution Date	

Test Case Execution :

Step No.	Test Steps	Input	Expected Result	Actual Result	Status



Testing Activities Performed :

Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable/Artifact
1				
2				
3				
4				
5				

Test Execution Report :



Test Case ID	Module Name	Test Scenario	Expected Result	Actual Result	Status (Pass/Fail)	Remarks

Test Summary Report :

Metric	Value
Total Test Cases Designed	
Total Test Cases Executed	
Total Passed Test Cases	
Total Failed Test Cases	
Total Blocked Test Cases	
Total Not Executed Test Cases	
Pass Percentage (%)	
Fail Percentage (%)	
Overall Test Status (Pass / Fail)	



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Tester Name	
Execution Date	

Observation :

Sr. No.	Observation	Remarks
1		
2		
3		
4		
5		

Output :



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Solve the below mention questionnaires:

1. What is Bugzilla primarily used for?

- A) Source Code Development
- B) Defect Tracking and Management
- C) Database Design
- D) UI Design

Answer:

2. What is TestLink primarily used for?

- A) Network Monitoring
- B) Test Case Management
- C) Database Testing
- D) Performance Monitoring



Answer:

3. Which status appears first in the Defect Life Cycle?

- A) Assigned
- B) Open
- C) New
- D) Closed

Answer:

4. Which document provides overall testing statistics and results?

- A) Requirement Document
- B) Design Document
- C) Test Summary Report
- D) User Manual

Answer:

5. What is the main benefit of using testing tools like Bugzilla and TestLink?

- A) Increase Coding Errors
- B) Reduce Software Features
- C) Improve Testing Efficiency and Traceability
- D) Eliminate Testing

Answer:

Conclusion / Outcome:

Rubrics:

Evaluation Criteria	Maximum Marks	Marks Obtained
Understanding of Defect Management, Test Case Management, and Bugzilla/TestLink Workflows	2	
Test Case Management and Defect Tracking Using Bugzilla/TestLink	3	
Test Execution and Report Generation	2	



Observation, Analysis, and Conclusion	2	
Documentation, Presentation, and Neatness	1	
Total	10	

Signature with Date of Completion	
Marks out of 10	

Date: _____

TUTORIAL -9 Black-Box & White-Box Testing

Aim: - Analyze Black-Box Testing, White-Box Testing, Integration Testing, and System Testing on a Sample Web Application.

Description :

Problem Statement :

Implementation Details :

Parameter	Details
Project Name	
Application/Software	



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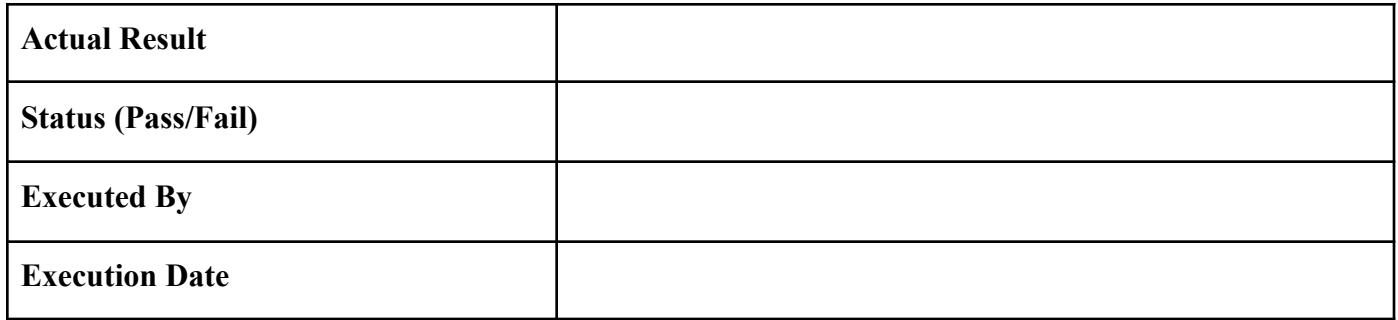
Module Name	
SDLC Model	
Programming Language	
IDE/Tool Used	
Input/Test Data	

Source Code :



Test Case Detailed :

Field	Details
Test Case ID	
Requirement ID	
Module Name	
Feature Under Test	
Test Scenario	
Test Objective	
Test Type (Unit/Integration/System/UAT)	
Test Technique (Black Box/White Box)	
Priority	
Severity	
Preconditions	
Postconditions	
Test Environment	
Browser/OS (if applicable)	
Test Data	
Expected Result	

[illegible]



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Testing Activities Performed:

Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable/Artifact

Observation :

Sr. No.	Observation	Remarks
1		
2		
3		
4		



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5		
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Output :



Solve the below mention questionnaires:

1. Which testing technique verifies functionality without examining source code?

- A) White-Box Testing
- B) Integration Testing
- C) Black-Box Testing
- D) Unit Testing

Answer:

2. Which testing technique requires knowledge of internal program code?

- A) System Testing
- B) Black-Box Testing
- C) Acceptance Testing
- D) White-Box Testing

Answer:

3. Integration Testing is mainly used to verify:

- A) Database Performance
- B) Communication between modules
- C) Source Code Formatting
- D) User Interface Design

Answer:

4. System Testing validates:

- A) Individual Functions Only
- B) Single Module Performance
- C) Complete Application Behavior
- D) Database Tables Only

Answer:

5. Which testing checks execution paths and branch coverage?



- A) Black-Box Testing
- B) White-Box Testing
- C) Regression Testing
- D) Usability Testing

Answer:

Conclusion / Outcome:

Rubrics:

Evaluation Criteria	Maximum Marks	Marks Obtained
Understanding of Black-Box, White-Box, Integration, and System Testing Concepts	2	
Analysis and Application of Testing Techniques on the Sample Web Application	3	
Test Cases, Test Execution, and Results	2	
Observation, Analysis, and Conclusion	2	
Documentation, Presentation, and Neatness	1	
Total	10	

Signature with Date of Completion	
Marks out of 10	

Date: _____

TUTORIAL -10 Automation & Regression Analysis

Aim: - Analyze black-box testing, white-box testing, integration testing, and system testing on a sample



web application.

Description :

Problem Statement :

Implementation Details :

Parameter	Details
Project Name	
Application/Software	
Module Name	
SDLC Model	
Programming Language	
IDE/Tool Used	
Input/Test Data	

Source Code :



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Module Analysis Table :

Module Name	Functionality	Critical (Yes/No)	Frequently Changed (Yes/No)	Suitable for Automation (Yes/No)	Reason



Automation Candidate Identification :

Test Case ID	Functionality	Automation Recommended (Yes/No)	Reason	Automation Tool

Test Case Detailed :

Field	Details
Test Case ID	
Requirement ID	
Module Name	



Feature Under Test	
Test Scenario	
Test Objective	
Test Type (Unit/Integration/System/UAT)	
Test Technique (Black Box/White Box)	
Priority	
Severity	
Preconditions	
Postconditions	
Test Environment	
Browser/OS (if applicable)	
Test Data	
Expected Result	
Actual Result	
Status (Pass/Fail)	
Executed By	
Execution Date	

Test Case Execution :

Step No.	Test Steps	Input	Expected Result	Actual Result	Status



Regression Test Suite :

Regression Test ID	Module	Scenario	Expected Result	Status

Automation Feasibility Report :

Criteria	Observation
Repetitive Test Cases	
Stable Functionality	
High Business Priority	



Frequently Executed	
Suitable Automation Tool	
Recommendation	

Regression Testing Summary :

Metric	Value
Total Modules Analyzed	
Automation Candidates	
Regression Test Cases	
Modules Excluded	
Recommendation	

Observation :

Sr. No.	Observation	Remarks



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Output :

Solve the below mention questionnaires:



1. Which testing technique verifies functionality without examining source code?

- A) White-Box Testing
- B) Integration Testing
- C) Black-Box Testing
- D) Unit Testing

Answer:

2. Which testing technique requires knowledge of internal program code?

- A) System Testing
- B) Black-Box Testing
- C) Acceptance Testing
- D) White-Box Testing

Answer:

3. Integration Testing is mainly used to verify:

- A) Database Performance
- B) Communication between modules
- C) Source Code Formatting
- D) User Interface Design

Answer:

4. System Testing validates:

- A) Individual Functions Only
- B) Single Module Performance
- C) Complete Application Behavior
- D) Database Tables Only

Answer:

5. Which testing checks execution paths and branch coverage?

- A) Black-Box Testing
- B) White-Box Testing
- C) Regression Testing
- D) Usability Testing

Answer:

Conclusion / Outcome:



Rubrics:

Evaluation Criteria	Maximum Marks	Marks Obtained
Understanding of Black-Box, White-Box, Integration, and System Testing Concepts	2	
Analysis and Implementation of Testing Techniques on the Sample Web Application	3	
Test Cases, Test Execution, and Test Results	2	
Observation, Analysis, and Conclusion	2	
Documentation, Presentation, and Neatness	1	
Total	10	

Signature with Date of Completion	
Marks out of 10	

Date: _____

TUTORIAL -11 Python Automation Testing

Aim: - Analyze automation testing in Python for website navigation, forms, multi-page workflows, links, and regression testing scenarios.

Description :

Problem Statement :



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Implementation Details :

Parameter	Details
Project Name	
Application/Software	
Module Name	
SDLC Model	
Programming Language	
IDE/Tool Used	
Input/Test Data	

Source Code :



SDLC–STLC Mapping Table

SDLC Phase	Corresponding STLC Phase	Testing Activity

Testing Activities Performed :



Sr. No.	SDLC Phase	STLC Phase	Testing Activity Performed	Deliverable / Artifact

Deliverables of Each Phase :

Sr. No.	SDLC Phase	Deliverable / Artifact	Purpose
1	Requirement Analysis		
2	System Design		
3	Development		
4	Testing		
5	Deployment		
6	Maintenance		

Roles & Responsibilities :

Sr. No.	Role	Responsibility	Phase Involved
1	Project Manager		



2	Business Analyst		
3	System Analyst		
4	Software Developer		
5	Software Tester / QA Engineer		
6	Database Administrator		
7	Client / End User		

Observation :

Sr. No.	Observation	Remarks
1		
2		
3		
4		
5		

Output :



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Solve the below mention questionnaires:

1. Which Python library is commonly used for web automation testing?

- A) NumPy
- B) Pandas
- C) Selenium
- D) Matplotlib

Answer:

2. Selenium WebDriver is mainly used for?

- A) Database Management
- B) Browser Automation
- C) Network Configuration
- D) Operating System Testing

Answer:



3. Which command installs Selenium in Python?

- A) install selenium
- B) python selenium
- C) pip install selenium
- D) selenium setup

Answer:

4. Regression Testing is performed to ensure that:

- A) New functionality is developed
- B) Existing functionality remains unaffected after changes
- C) Database is created
- D) Source code is deleted

Answer:

5. Which Selenium component is used for browser automation?

- A) Selenium Grid
- B) Selenium IDE
- C) Selenium WebDriver
- D) Selenium Server

Answer:

Conclusion / Outcome:

Rubrics:

Evaluation Criteria	Maximum Marks	Marks Obtained
Understanding of Automation Testing Concepts and Python Testing Framework	2	
Automation Script Development for Website Navigation, Forms, and Multi-Page Workflows	3	
Test Execution, Regression Testing, and Result Analysis	2	
Observation, Analysis, and Conclusion	2	



Documentation, Presentation, and Neatness	1	
Total	10	

Signature with Date of Completion	
Marks out of 10	

Date: _____

TUTORIAL -12 AI-Powered Software Testing

Aim: - Analyze AI-Powered Software Testing Tools and Apply Generative AI Techniques for Test Case Generation, Test Data Creation, Defect Prediction, Requirement Analysis, and Automated Selenium Test Script Generation for Software Quality Assurance.

Description :

Problem Statement :

Implementation Details :

Parameter	Details
Project Name	
Application/Software	



Module Name	
SDLC Model	
Programming Language	
IDE/Tool Used	
Input/Test Data	

Automation Test Configuration :

Parameter	Details
Automation Tool	
Python Version	
Selenium Version	
WebDriver Used	
Browser	
Target Website URL	
Test Framework	
Execution Date	

Source Code :



Automation Test Case Details :

Field	Details
Test Case ID	
Module Name	
Automation Script Name	
Test Scenario	
Test Objective	
Test Data	
Expected Result	
Actual Result	



Status	
Executed By	
Execution Date	

Automation Test Execution :

Step No.	Automation Action	Expected Result	Actual Result	Status

Regression Test Execution Report :

Regression Test ID	Module	Scenario	Expected Result	Actual Result	Status



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Automation Execution Summary :

Metric	Value
Total Automation Scripts	
Scripts Executed	
Passed	
Failed	
Regression Tests Executed	
Execution Time	
Overall Status	

Observation :

Sr. No.	Observation	Remarks



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Output :



Solve the below mention questionnaires:

1. Which technology is primarily used in this practical way to automate web browsers?

- A) NumPy
- B) Pandas
- C) Selenium WebDriver
- D) Matplotlib

Answer:

2. What is the main advantage of using Generative AI for test case generation?

- A) It removes the need for testing completely.
- B) It automatically deploys applications.
- C) It speeds up creation of test cases from requirements.
- D) It replaces browsers during automation.

Answer:

3. Which testing artifact can Generative AI create from a login requirement?

- A) Database schema
- B) Hardware driver
- C) Selenium automation script
- D) Network router configuration

Answer:

4. Defect prediction is mainly used to:

- A) Replace source code review
- B) Identify modules likely to contain defects
- C) Generate UI designs
- D) Create database indexes

Answer:



5. Why should AI-generated test cases and scripts still be reviewed by humans?

- A) AI cannot generate any code.
- B) Browsers do not support Selenium.
- C) Generated artifacts may contain errors, gaps, or incorrect assumptions.
- D) Test cases do not require validation.

Answer:

Conclusion / Outcome:

Rubrics:

Evaluation Criteria	Maximum Marks	Marks Obtained
Understanding of AI-Powered Software Testing Tools and Generative AI Concepts	2	
Application of AI Techniques for Test Case Generation, Test Data Creation, Requirement Analysis, and Defect Prediction	3	
Automated Selenium Test Script Generation, Execution, and Result Analysis	2	
Observation, Analysis, and Conclusion	2	
Documentation, Presentation, and Neatness	1	
Total	10	

Signature with Date of Completion	
Marks out of 10	

Date: _____



TUTORIAL -13 Selenium Environment Setup

Aim: - Create and Configure Python, Selenium WebDriver, VS Code/PyCharm IDE, Browser Drivers, and Automation Testing Libraries for Selenium Testing.

Description :

Problem Statement :

Implementation Details :

Parameter	Details
Project Name	
Application/Software	
Module Name	
SDLC Model	
Programming Language	
IDE/Tool Used	
Input/Test Data	

SDLC–STLC Mapping Table :



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SDLC Phase	Corresponding STLC Phase	Objective	Key Activities

Source Code :



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Test Case Detailed :

Field	Details
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Test Case ID	
Requirement ID	
Module Name	
Feature Under Test	
Test Scenario	
Test Objective	
Test Type (Unit/Integration/System/UAT)	
Test Technique (Black Box/White Box)	
Priority	
Severity	
Preconditions	
Postconditions	
Test Environment	
Browser/OS (if applicable)	
Test Data	
Expected Result	
Actual Result	
Status (Pass/Fail)	
Executed By	
Execution Date	

Test Case Execution :

Step No.	Test Steps	Input	Expected Result	Actual Result	Status
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[illegible]

Testing Activities Performed

Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable/Artifact
1				



2				
3				
4				
5				

Project Deliverables Table :

SDLC Phase	Deliverable / Artifact	Prepared By
Requirement Analysis		
System Design		
Development		
Testing		
Deployment		
Maintenance		

Roles & Responsibilities :

Role	Responsibility	SDLC/STLC Phase
Project Manager		
Business Analyst		



Software Developer		
Software Tester		
Client / End User		

Project Documents / Artifacts Checklist :

Sr. No.	Document / Artifact	Purpose	Status (✓ / X)
1	Software Requirement Specification (SRS)	Defines software requirements	
2	Test Plan	Planning testing activities	
3	Test Cases	Defines test scenarios	
4	Test Data	Input used for testing	
5	Bug Report	Records identified defects	
6	Test Summary Report	Summarizes testing results	

Observation :

Sr. No.	Observation	Remarks
1		
2		



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3		
4		
5		

Output :



Solve the below mention questionnaires:

1. Which library is used for browser automation in Python?

- A) NumPy
- B) Pandas
- C) Selenium
- D) Matplotlib

Answer:

2. What is the purpose of ChromeDriver?

- A) Database Management
- B) Browser Communication with Selenium
- C) Code Compilation
- D) Report Generation

Answer:

3. Which command is used to install Selenium?

- A) install selenium
- B) python selenium
- C) pip install selenium
- D) setup selenium

Answer:

4. Which IDE is commonly used for Selenium script development?

- A) MS Paint
- B) VS Code
- C) Calculator
- D) Notepad Only

Answer:

5. What is the purpose of WebDriver Manager?

- A) Database Management
- B) Driver Download and Configuration Automation
- C) Browser Removal
- D) Network Monitoring

Answer:

Conclusion / Outcome:



Rubrics:

Evaluation Criteria	Maximum Marks	Marks Obtained
Understanding of Python, Selenium WebDriver, IDEs, and Automation Testing Environment	2	
Installation and Configuration of Python, Selenium WebDriver, Browser Drivers, and Required Libraries	3	
IDE Setup, Environment Verification, and Successful Execution of a Sample Selenium Script	2	
Observation, Analysis, and Conclusion	2	
Documentation, Presentation, and Neatness	1	
Total	10	

Signature with Date of Completion	
Marks out of 10	

Date: _____

TUTORIAL -14 Selenium Web Automation

Aim: - Create Selenium Automation Scripts in Python for Browser Navigation, Form Handling, Web Element Interaction, and Validation Testing.

Description :



Problem Statement :

Implementation Details :

Parameter	Details
Project Name	
Application/Software	
Module Name	
SDLC Model	
Programming Language	
IDE/Tool Used	
Input/Test Data	

Automation Environment Configuration :

Parameter	Details
Python Version	
Selenium Version	
Browser	
WebDriver	
IDE Used	
Target Website URL	

Automation Scenario Table :



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Scenario ID	Scenario	Module/Page	Expected Result
1	Browser Navigation		
2	Form Handling		
3	Web Element Interaction		
4	Validation Testing		

Source Code :



Automation Test Case Details :

Field	Details
Test Case ID	
Script Name	
Module Name	
Test Scenario	
Test Data	
Expected Result	
Actual Result	
Status (Pass/Fail)	

Automation Script Execution :



Step No.	Automation Action	Expected Result	Actual Result	Status

Web Element Identification Table :

Element Name	Locator Type	Locator Value	Purpose

Validation Results :



Validation	Expected Result	Actual Result	Status
Page Title			
URL			
Form Submission			
Success/Error Message			

Automation Execution Summary :

Metric	Value
Total Test Scripts	
Executed	
Passed	
Failed	
Execution Time	

Observation :

Sr. No.	Observation	Remarks



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Output :



olve the below mention questionnaires:

Q1. What is Selenium?

Answer:

Q2. What is Selenium WebDriver?

Answer:

Q3. What is ChromeDriver?

Answer:

Q4. Which method is used to enter text into a textbox?

Answer:

Q5. Which method is used to click a button?

Answer:

Conclusion / Outcome:

Rubrics:

Evaluation Criteria	Maximum Marks	Marks Obtained
Understanding of Selenium Automation, Browser Navigation, and Web Element Interaction	2	
Development of Selenium Automation Scripts for Browser Navigation and Form Handling	3	
Validation Testing, Script Execution, and Result Verification	2	
Observation, Analysis, and Conclusion	2	



Documentation, Presentation, and Neatness	1	
Total	10	

Signature with Date of Completion	
Marks out of 10	

Date: _____

TUTORIAL -15 Automated Login Testing

Aim: -Create automated login functionality, authentication validation, assertions, and exception handling scenarios using Selenium WebDriver with Python.

Description :

Problem Statement :

Implementation Details :

Parameter	Details
Project Name	
Application/Software	
Module Name	
SDLC Model	



Programming Language	
IDE/Tool Used	
Input/Test Data	

Automation Environment Configuration :

Parameter	Details
Python Version	
Selenium Version	
Browser	
WebDriver	
IDE Used	
Target URL	

Login Automation Scenario :

Scenario ID	Scenario	Expected Result
TC-01	Valid Login	
TC-02	Invalid Password	
TC-03	Invalid Username	



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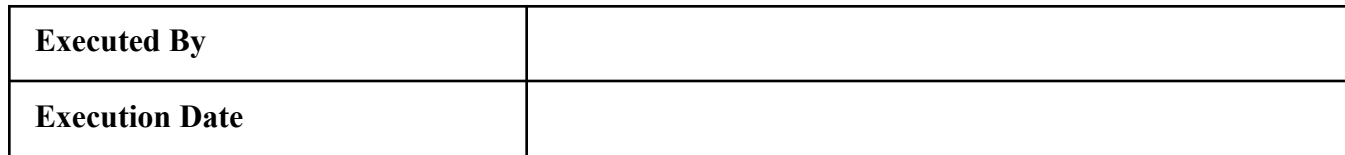
TC-04	Empty Credentials	
TC-05	Logout	

Source Code :



Test Case Detailed :

Field	Details
Test Case ID	
Requirement ID	
Module Name	
Feature Under Test	
Test Scenario	
Test Objective	
Test Type (Unit/Integration/System/UAT)	
Test Technique (Black Box/White Box)	
Priority	
Severity	
Preconditions	
Postconditions	
Test Environment	
Browser/OS (if applicable)	
Test Data	
Expected Result	
Actual Result	
Status (Pass/Fail)	

[illegible]



Web Element Identification Table :

Element Name	Locator Type	Locator Value	Purpose
Username			
Password			
Login Button			
Error Message			
Logout Button			

Assertion Validation Report :

Assertion	Expected Result	Actual Result	Status
Page Title			
URL			
Welcome Message			
Login Success			
Error Message			

Exception Handling Report :



Exception	Possible Cause	Handled (Yes/No)	Remarks
NoSuchElementException			
TimeoutException			
ElementNotInteractableException			
StaleElementReferenceException			

Automation Execution Summary :

Metric	Value
Total Test Cases	
Executed	
Passed	
Failed	
Assertions Passed	
Exceptions Handled	

Observation :

Sr. No.	Observation	Remarks
1		
2		
3		
4		
5		

Output :



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Solve the below mention questionnaires:

Q1. What is Authentication Testing?

Answer:

Q2. What is Selenium WebDriver?

Answer:

Q3. What is Assertion?

Answer:



Q4. What is Exception Handling?

Answer:

Q5. Which exception occurs when an element cannot be found?

Answer:

Conclusion / Outcome:

Rubrics:

Evaluation Criteria	Maximum Marks	Marks Obtained
Understanding of Automated Login, Authentication Validation, Assertions, and Exception Handling	2	
Development of Selenium WebDriver Scripts for Login and Authentication Testing	3	
Assertions, Exception Handling, Script Execution, and Result Verification	2	
Observation, Analysis, and Conclusion	2	
Documentation, Presentation, and Neatness	1	
Total	10	

Signature with Date of Completion	
Marks out of 10	